Sahaj Bhadja

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Summary

Software engineer with 2+ years building reliable systems using **C, C++, and Python**, including autonomous-driving research, closed-loop simulation, low-latency APIs, concurrency, automated testing, Linux, CI/CD, and cross-functional technical delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Developed scalable Python software from technical requirements through architecture, implementation, testing, and deployment, processing 5,000+ monthly workflows with failure isolation, error handling, and 99%+ production reliability.
- Optimized critical concurrent backend services and database interfaces for 500+ users, reducing latency from 1.2 seconds to 350 milliseconds through profiling, algorithm improvement, validation, and targeted debugging.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built clean, well-tested Python services handling 10,000+ daily requests, implementing scalable multithreaded batch processing, secure ingestion, and fault-tolerant execution while maintaining reliable performance for 150+ instructors.
- Drove complex features from requirements through implementation and automated testing, clearly communicating technical tradeoffs across teams while reducing feedback turnaround 60% and sustaining 90%+ model evaluation consistency.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built large-scale autonomous-driving simulation pipelines processing 100,000+ trajectory logs, improving long-horizon planning 18% through Python model development, algorithm optimization, and rigorous, reproducible closed-loop system performance evaluation.
- Designed a spatial-temporal architecture integrating road geometry, lane direction, ego state, and vehicle trajectories, validating 10,000 scenarios and improving closed-loop planning stability 15% while reducing off-road behavior.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered cross-platform software using complex object-oriented patterns and 20+ automated unit and integration tests across Linux, Windows, and macOS, reducing technical debt 20% and response times 30%.
- Debugged complex distributed API and integrated data-processing issues, optimizing low-bandwidth execution with AWS Glue and S3 while improving pipeline efficiency 68% through coordinated Agile delivery and CI/CD.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Implemented eight modular services for real-time resource allocation, customer workflows, billing, permissions, and notifications, synchronizing two million records across locations through REST APIs, WebSockets, and Azure SQL.
- Automated unit, integration, and end-to-end testing through reliable CI/CD pipelines and safe staged deployments, achieving 95%+ build validation while reducing deployment time 50% and code-review turnaround 25%.

Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, SQL, Shell, Bash
Autonomy: Autonomous systems, trajectory planning, behavioral planning, closed-loop simulation, HD-map geometry, offline reinforcement learning
Systems: Linux, Unix, concurrency, multithreading, distributed systems, microservices, REST APIs, WebSockets, low-latency optimization
Testing: Unit testing, integration testing, system testing, regression testing, validation, simulation, debugging, performance profiling
Infrastructure: Git, GitHub Actions, Jenkins, Docker, Kubernetes, CI/CD, PyTest, AWS, technical design, Agile, technical documentation

Projects

CPU Scheduling Simulator Live | *C++, Algorithms, Simulation*

- Implemented six C++ scheduling algorithms with edge-case handling, interactive simulation, and Gantt visualization, validating 1,000+ process combinations while reducing average turnaround time 25% and debugging time 20%.

Linguistix | *Python, AWS, CI/CD*

- Developed a containerized Python service with reliable automated build, validation, and deployment infrastructure, achieving 87% measured accuracy and dependable production operation through repeatable CI/CD-based verification on AWS.

CaptionCraft Live | *Python, PyTorch, Test Infrastructure*

- Designed repeatable test infrastructure across four software models, 10,000 images, and 100,000 outputs, analyzing failures and performance tradeoffs to improve measured system accuracy from 81% to 96%.